

Coexistence of *bla*_{NDM-1}, *bla*_{OXA-51}, *bla*_{OXA-23}, and *armA* in conjunction with novel mutations detected in RND efflux pump regulators in tigecycline resistant clinical isolates of *Acinetobacter baumannii*

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One sentence summary: Novel mutations (deletions/substitutions) detected in efflux pump regulatory genes were associated with increased levels of tigecycline resistance.

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Abstract

This study has investigated a total of 51 *Acinetobacter baumannii* isolates for the prevalence of resistant determinants in tigecycline susceptible and non-susceptible clinical isolates of *A. baumannii*. Antimicrobial susceptibility testing revealed 74% of isolates were tigecycline resistant. Mutations in RND-efflux pump regulatory genes and the expression of efflux pump genes were measured in tigecycline resistant isolates. There was a strong co-relation between the *bla*_{NDM-1} and *armA* wherein majority of the isolates that are positive for *bla*_{NDM-1} have also harbored *armA*. Compared with TSAB (tigecycline susceptible *A. baumannii*), TNAB (tigecycline non-susceptible *A. baumannii*) isolates show increased distribution of *bla*_{NDM-1} ($P = 0.048$), *bla*_{IMP-1} ($P < 0.0001$) and *bla*_{OXA-51} ($P = 0.0029$) carbapenemase genes. The variants of RND-efflux pump regulatory genes due to amino-acid mutations in *adeS* (F12S, K84E, W61R, N268H and Q299R) and *adeL* (G21R and Q262R) were identified in tigecycline resistant isolates as well as ISAbal mediated disruption of *adeN* were observed causing overexpression of *adeIJK* efflux pump. Additionally, mutations in *adeRS* were also associated with increased expression of *adeABC* efflux pump. Besides, TNAB isolates showed significantly ($P < 0.0001$) higher ability of biofilm formation as compared to TSAB isolates. The tigecycline resistance due to mutations in contemporary *A. baumannii* isolates having a higher ability to form biofilm may pose therapeutic difficulties.

Keywords: tigecycline resistance, RND-efflux pump regulator, novel mutations, biofilms, *bla*_{NDM-1}

Introduction

Acinetobacter baumannii infection is a serious problem among immunocompromised patients in ICU and their resistance to different classes of antibiotics makes the treatment still more difficult (Lee et al. 2017). One recent study performed in Indian hospitals revealed that infections caused by *Acinetobacter* species were most common among all the nosocomial infections encountered (DeNIS 2016). One recent surveillance study performed in Europe revealed that over 70% of all invasive *Acinetobacter* isolates are resistant to carbapenems and most of them were isolated from older patients admitted to ICUs (Ayobami et al. 2020). The resistance in *A. baumannii* is can be intrinsic by chromosomally encoding genes or by plasmid encoding resistance determinants. *A. baumannii* is known to have the capacity of acquiring resistance through horizontal gene transfer (HGT) and via mobile genetic elements (Da Silva and Domingues 2016, Cooper et al. 2017). In some of the *A. baumannii* isolates, all the resistance determinants responsible for developing resistance towards most of the antimicrobial agents were seen to be clustered in a specific region of the genome referred as Resistance Island (Sepu-

tiene et al. 2012). The carbapenem hydrolyzing enzymes produced by *bla*_{OXA-51}, *bla*_{OXA-23}, and *bla*_{OXA-58} make carbapenems ineffective against *A. baumannii* (Ramadan et al. 2018). Carbapenem resistance in *A. baumannii* is commonly associated with the presence of Metallo-β-lactamases (MBL) such as *bla*_{NDM-1} (Krizova et al. 2012). The main MBLs found in *A. baumannii* are *bla*_{NDM-1}, *bla*_{IMP-1}, *bla*_{SIM-1} and *bla*_{VIM} that are responsible for carbapenem resistance (Girija et al. 2018, Kaur and Singh 2018). The high prevalence of carbapenemase-producing MDR *A. baumannii* is reported in different parts of the world including lower-middle-income countries like India, Nepal, Pakistan, Bangladesh etc. (Gandra et al. 2020). An increase in carbapenem resistance in *A. baumannii* is directly associated with an increased mortality rate (Lemos et al. 2014).

Resistance to aminoglycosides is due to enzymatic modification by Aminoglycoside Modifying Enzymes (AMEs) and 16S rRNA methyltransferases and the presence of these genes is responsible for high-level of aminoglycoside resistance among *A. baumannii* isolates (Nie et al. 2014). As *A. baumannii* resistance towards carbapenems and aminoglycosides is now a frequent phenomenon, colistin and tigecycline are considered the last re-

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sort drugs against *A. baumannii* infections. Due to the risk of nephrotoxicity and the development of hetero-resistance, there are limitations for using these antibiotics (Michalopoulos and Karatza 2010, Karaaslan et al. 2016). The tigecycline resistance in *A. baumannii* may cause due to amino acid substitutions in efflux pump regulatory genes, (Yoon et al. 2013) TetA (flavin-dependent monooxygenase) (Moore et al. 2005), TetX (Wang et al. 2019, Liu et al. 2020), mutations in 1-acyl-sn-glycerol-3-phosphate acyl-transferase (plsC) (Li et al. 2015) and tigecycline-related methyl-transferase (trm)(Chen et al., 2014). Other resistance mechanisms in *A. baumannii* involve resistance-nodulation-cell division (RND) type efflux pumps such as AdeABC, AdeFGH, and AdeIJK. These RND efflux pumps are associated with tigecycline and carbapenem resistance due to altered expression levels of efflux pump genes (Coyne et al. 2011, Li et al. 2015). The carbapenem resistance is very common in Indian clinical isolates of *A. baumannii*, thus tigecycline and polymyxin are the next best available treatment options. Since tigecycline resistance is also slowly emerging in *Acinetobacter*, the present study intended to evaluate the current resistant trends for tigecycline among recent *A. baumannii* isolates of Indian origin. Further, the study aims to investigate the tigecycline resistant mechanisms in these *A. baumannii* isolates.

Material and Methods

Bacterial isolates and species identification

A total of 51 clinical isolates of *A. baumannii* were obtained from tertiary care referral hospital, Pondicherry Institute of Medical Sciences (PIMS), Puducherry, India, during the period 2017–2018. *A. baumannii* isolates were collected from various clinical specimens, such as endotracheal aspirates ($n = 20$), wounds ($n = 7$), blood ($n = 15$), bronchoalveolar lavage ($n = 2$), pus ($n = 3$), tissue ($n = 3$) and sputum ($n = 1$). Species identification was confirmed by PCR detection of *bla*_{OXA-51-like} carbapenemase gene that is intrinsic to *A. baumannii*. Further, it was confirmed through 16S rRNA gene amplification and its sequence analysis followed by BLAST analysis.

REP-PCR analysis

All the isolates of *A. baumannii* were subjected to genotyping by repetitive extragenic palindromic polymerase chain reaction (REP-PCR) with the primer (REP-F-CGGAATGGCTCATCAGATC and REP-R-IIIGCGCCGICATCAGGC) used in the previous study (Martín-Lozano et al. 2002). The PCR reaction mixture contained Taq DNA Polymerase Master Mix RED (Ampliqon, Denmark) 10 μ M of each primer, 10 ng DNA template. PCR amplification was performed in Applied Biosystems Veriti® 96-Well Thermal Cycler and analyzed by electrophoresis (1% agarose) and its documentation was done through a gel documentation system (Bio-Rad) and comparative analysis of the amplicon profiles were performed using PyElph neighbor-joining method (Pavel and Vasile 2012).

Antimicrobial susceptibility testing

Minimum inhibitory concentration (MIC) for all the isolates was determined using the micro-broth dilution method in MH broth following the current CLSI guidelines (CLSI 2020). The MIC values (mg/L) of tigecycline and colistin were determined by both the broth microdilution method and the E-test (Hi-media, India). MIC for tigecycline resistance is defined as (MIC > 0.5 mg/L) same as *Enterobacteriaceae* according to EUCAST, as there are no breakpoints specified for *A. baumannii* (EUCAST 2020).

Biofilm formation

The biofilm-forming ability of the *A. baumannii* isolates was tested quantitatively using the microtiter plate assay (Rao et al. 2008). Briefly, strains were grown overnight at 37°C in brain heart infusion (BHI) broth. In each well of 96-well microtitre plate, 198 μ l of sterile BHI broth was added and 2 μ l of each of the overnight grown cultures were inoculated into the wells in triplicates followed by 24 h incubation without shaking at 37°C. The biofilm was quantified by optical density measurements using the standard protocol (Rao et al. 2008).

PCR-based methods to detect the presence of resistance determinants

Crude cell lysates or genomic DNA were used as templates for PCR. PCR assays were performed to detect *bla*_{OXA-51}, *bla*_{OXA23}, *bla*_{OXA-58}, *bla*_{PER-1}, *bla*_{NDM-1}, *bla*_{IMP-1}, *bla*_{ADC}, *armA*, and efflux pump genes as described previously (Saranathan et al. 2017a). All primers used to detect various β -lactamases, AMEs, insertion elements (IE), and ISAbal in *adeN* gene were listed in supplementary table—Table S1. All PCR assays performed used either 16S rRNA or *bla*_{OXA-51} gene as the internal control. The disruption in *adeN* by ISAbal was determined by witnessing an increase in PCR amplicon size than expected. A second nested PCR followed by earlier PCR using primers of ISAbal was performed for confirming disruption. *A. baumannii* ATCC17978 and PKAB07 (Saranathan et al. 2014) strains were used as a negative and positive control respectively. Representative PCR products of each resistant determinant were confirmed by Sanger sequencing (Macrogen, Korea).

Nucleotide sequencing and data analysis

To investigate the tigecycline resistance mechanisms in *A. baumannii* isolates, amino acid sequences of efflux pump regulatory genes (*adeN* TetR type regulator of *adeIJK*, *adeL* LysR type regulator of *adeFGH* and *adeRS* two-component system of *AdeABC*) from randomly selected tigecycline resistant *A. baumannii* clinical isolates were compared with the sequences of the reference strain *A. baumannii* ATCC17978. Full-length RND efflux pump genes were amplified using primers (Table S1). The amplicons were purified with GeneJET PCR purification kit (Thermo Scientific, USA), and Sanger sequencing was done at Macrogen Inc, South Korea. The sequence was analyzed BLAST program searches were performed using the National Center for Biotechnology Information website (<http://www.ncbi.nlm.nih.gov>). Multiple sequence alignment of the sequences was done using the program MEGA Version X and Clustal Omega server (<https://www.ebi.ac.uk/Tools/msa/clustalo/>).

qRT-PCR

The relative expression of *adeG*, *adeB* and *adeJ* was determined in Tigecycline Susceptible *A. baumannii* (TSAB) and Tigecycline Non-susceptible *A. baumannii* (TNSAB) isolates wherein representative isolates from each group was tested and compared. The expression level of *adeJ* was used to determine the effect of disruption of TetR regulator *adeN* due ISAbal insertion on the expression of *adeIJK* efflux pump as per our earlier investigation (Saranathan et al. 2017b) and these results will be correlated with the tigecycline resistance. Total RNA was extracted from cultures grown in LB until mid-log phase using Trizol reagent (Sigma-Aldrich, Germany) Total RNA was treated with RNase-free DNase (Thermo Scientific, USA) to remove any cellular DNA. RNA was reverse transcribed into cDNA using first strand cDNA synthesis kit (Thermo Scientific, USA). qPCR was performed with SYBR Green master mix Roche with ROX (Roche, USA) using Roche Light Cycler® 96 Sys-

tem with the following thermal steps: 95°C for 10 min followed by 40 cycles of 95°C for 10 s and 58°C for 30 sec and 72°C for 30 Sec. All the qPCR analysis was done with three replicates, housekeeping gene *rpoB* was used as an internal reference gene to quantify the transcription. The data were analyzed by the relative expression rate that was measured by the $2^{-\Delta\Delta CT}$ method (Schmittgen and Livak 2008).

Construction of Knockouts

The knockout of gene encoding *adeR* response regulator and *adeS* histidine kinase of the *A. baumannii* ATCC 17978 strain was created by allelic replacement according to the published procedure (Hamad et al. 2009). The flanking regions (1 Kb each) of *adeR* and *adeS* from *A. baumannii* ATCC 17978 strain were amplified by PCR and cloned into the suicide vector pMo130-Tel^R (Table 1) in *E. coli* DH5 α competent cells. The plasmid (pMo130-Tel^R) containing the forward and reverse flanking regions were subsequently moved to *A. baumannii* ATCC 17978 strain by electroporation. pMo130-Tel^R with the flanking region of respective genes integrated into the genome of *A. baumannii* ATCC 17978 strain by homologous recombination (first crossovers) were selected on LB agar containing 50 μ g/ml Kanamycin. Transconjugants were cultured in LB broth containing 10% sucrose to select double recombinants, and serial dilutions were spread onto LB plates containing 10% sucrose. Visual detection of XylE activity following catechol treatment was also used to differentiate between first and second recombinants after growing cells on sucrose-containing media. The deletion *adeR* and *adeS* were confirmed by PCR amplification of primers that anneal at flanking regions (Fig. S2A).

Plasmid and strain construction/Generation of the shuttle vector

The *E. coli* cloning plasmid pUC19 was modified for cloning in *A. baumannii* by cloning the 1.3 kb ori (origin of replication) from the pWH1266 shuttle vector specific for *Acinetobacter*. The Kanamycin resistance marker was amplified from the pET28a expression vector and cloned in the pUC19: ori-containing plasmid (Fig. S2B). The resultant plasmid was designated as pUCAB and was further used for the cloning of *adeR* and *adeS* wild type (WT) genes from *A. baumannii* ATCC 17978 strain.

Complementation using pUCAB

Intact full-length *adeS* were amplified from the *A. baumannii* ATCC 17978 genomic DNA using primers (Table S1). The amplicons were purified using a GeneJET PCR clean-up kit. Purified amplicons and modified plasmid pUCAB were digested using *Bam*HI and *Sma*I (Thermo scientific, USA) at 37°C and ligation reactions were performed using T4 DNA ligase (Thermo Scientific, USA) at 22°C. The ligation mixture was then used to transform *E. coli* DH5 α chemical competent cells by heat shock and plated on LB agar containing 50 μ g/ml kanamycin. The transformants were screened by PCR and the plasmid was isolated for insert release from the positive colonies. The plasmid with insert was moved to respective knockout cells (Δ *adeS*) of *A. baumannii* by electroporation and clones were selected on LB agar with 50 μ g/ml Kanamycin and further confirmed by PCR.

Introduction of amino acid substitutions by Site-directed mutagenesis

The PCR-based method was employed to exchange a single nucleotide in the sequence of *adeS* (Jeltsch and Lanio 2002). The complement plasmid having WT-*adeS* was used as a template in order

to introduce amino acid substitution F12S in *adeS*. Briefly, the amplicon of 160 bp of *adeS* was amplified with the primer having desired nucleotide change in order to bring amino acid change. The plasmid with WT *adeS* was used as a template for rolling circle amplification while a 160 bp amplicon was used as a primer. The resultant PCR product was purified and digested with the *Dpn*I enzyme to remove the template plasmid. The resultant product was used to transform the *E. coli* DH5 α cells by heat shock method. The transformants were selected on Kanamycin followed by plasmid isolation. The plasmid was then moved to the Δ *adeS* knockout strain of *A. baumannii*. Sanger sequencing was used to confirm the correct nucleotide exchange.

Nucleotide sequence accession number

Total 12 nucleotide sequences of RND-efflux pump regulators having novel mutations and disruption due to *ISAb*a1 insertion sequence was submitted to GenBank with accession number from **MZ542334 to MZ542345**.

Result and Discussion

PCR screening for β -Lactamases encoding genes

PCR screening for common MBL genes *bla*_{NDM-1} and *bla*_{IMP-1} showed 11(21%) isolates were harboring the *bla*_{NDM-1}, while 24 (47%) isolates were found positive for *bla*_{IMP-1}. In total, around 21 (41%) isolates showed to have either of *bla*_{NDM-1} or *bla*_{IMP-1} or both of MBL genes. Only one isolate found positive for *bla*_{OXA-58}, which is very rare variant observed among Indian isolates. A total of 41 isolates (80%) were found have *bla*_{OXA-23} and this resistant determinant showed maximum prevalence among all the β -lactamases genes screened. Therefore, the major genotypic determinant for drug resistance is oxacillinase *bla*_{OXA-23} among the isolates of Indian origin. Ambler class β -lactamase *bla*_{PER-1} was seen only in 19 isolates (37%). *Acinetobacter*-derived cephalosporinase *bla*_{ADC} gene was found in 33 isolates (64%), which is significantly lesser than in the previous investigation (Saranathan et al. 2017a) wherein the prevalence of 100% was observed among the tested isolates.

AMEs and 16S rRNA methyltransferases

A total of 39 (76%) isolates were found positive for the methyltransferase *armA*, while none of the isolates had other methyltransferase genes (*rmtA*, *rmtB*, *rmtC* and *rmtD*) that were tested. As all the isolates were screened for common AMEs such as *aac31*, *ant31*, *aph31*, and *aph61*, PCR results revealed that occurrence of common AMEs in varying percentages wherein 33 isolates (65%) was found positive for *aph31*, which is the maximum, while only 9 isolates harbored *aac31* (17%). Other AMEs such as *ant31* and *aph6b* were found in 23 (45%) and 20 (39%) isolates respectively. A total of 37 isolates (72%) encoded at least one of the AMEs tested. There is a paucity of studies with respect to the detection of AMEs and 16S rRNA methyltransferases in this region.

Antimicrobial susceptibility testing

All the isolates were resistant to ceftazidime, ciprofloxacin and chloramphenicol while the least resistance was found for colistin where only two isolates (4%) were resistant to colistin. The colistin-resistant isolates were also resistant to all other antibiotics tested indicating the emergence of pan drug resistance. Among a total of 51 isolates, 50 (98%) were resistant for tetracycline, 48 isolates were resistant to aminoglycosides, around 48 (94%), 45 (88%) and 43 (84%) isolates were resistant to tobramycin, amikacin and gentamicin respectively. A total of 78.72% of iso-

Table 1. Bacterial strains and plasmids used for the generation of knockout and site-directed mutagenesis experiment.

Strain/plasmid	Characteristics	Reference
<i>E. coli</i> DH5 α	Competent cells for cloning	Laboratory stock
pUC19	Plasmid backbone	Laboratory stock
pMO130-Tel ^R	Suicidal vector (Kan ^R Tel ^R)	Addgene
pWH1266	Shuttle vector (Amp ^R)	Saranaathan et al., 2017b
pUCAB	Kan ^R Shuttle vector	This study
<i>A. baumannii</i> ATCC17978	WT reference strain	ATCC
Δ adeR	adeR deleted in WT strain	This study
Δ adeS	adeS deleted in WT strain	This study
Δ adeS::pUCAB-adeS (CP)	adeS complement	This study
Δ adeS::pUCAB-adeS(F12S)	adeS (F12S) complement	This study

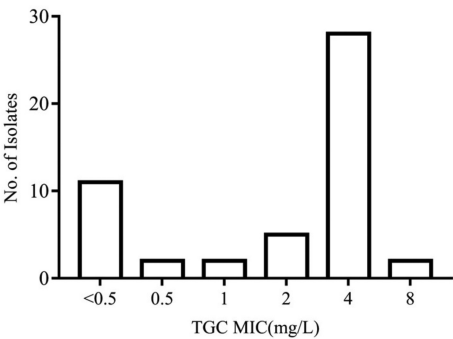


Figure 1. Tigecycline MIC distribution for 51 investigated *A. baumannii* isolates.

lates showed a high level of aminoglycoside resistance for all the three aminoglycoside antibiotics tested wherein their MIC levels were ≥ 256 mg/L. Tigecycline resistance was observed in the significant number of isolates wherein around 74.5% of isolates ($n = 38$) were resistant to tigecycline. This is the first report from India describing such high level of resistance to tigecycline.

Besides carbapenems, tigecycline, and polymyxins were also used for treating *A. baumannii* infections. Tigecycline is a broad-spectrum tetracycline class of antibiotic used against most of the MDR bacterial infections. According to earlier reports, resistance to tigecycline in *A. baumannii* is associated with overexpression of RND efflux pumps mainly AdeABC, AdeFGH, and AdeIJK (Ardehali et al. 2019). Genetic modifications (due to deletion mutation in tigecycline-related methyltransferase (*trm*) and frame shift mutation in *plsC* gene were also held responsible for decrease in susceptibility to tigecycline (Coyne et al. 2011), however, we did not find any such mutations in our study. Each of the RND efflux pump operons is regulated by different transcriptional regulators, for example, *adeIJK* is regulated by Tet-R type transcription factor called *adeN* (Rosenfeld et al. 2012) and the AdeABC efflux pump is regulated by two component system called *adeRS* (Coyne et al. 2011, Colclough et al. 2020). In this study, a total 38 isolates (74.50%) were showing resistance towards tigecycline with the MIC levels of > 0.5 (Fig. 1). The antibiotic resistance pattern and resistant determinants carried by TNSAB are shown in Fig. 2. In order to find out the mechanism of tigecycline resistance, while performing MIC assays we have used cyanide 3-chlorophenylhydrazine (CCCP) along with tigecycline, an uncoupler of oxidative phosphorylation, which can disrupt the proton gradient of the bacterial membranes thereby inhibiting the functions of RND efflux pumps (Park and Ko 2015). This assay results revealed that the MIC of tigecycline in presence of CCCP was decreased more than two fold.

Before testing susceptibility in combination with efflux pump inhibitor (EPI) CCCP the MIC for CCCP alone was tested, which was 100 mg/L for all the isolates. All the tigecycline resistant isolates were grown in presence of 4 mg/l of CCCP along with tigecycline to study the role of efflux pump in tigecycline resistance. The effects of the EPIs in combination with tigecycline MIC is shown in Table 2. Forty-nine isolates (96%) classified under both TNSAB and TSAB were showing decrease in resistance ranging from two to eight-fold. Only two isolates (3.9%) showed no change or reduction in tigecycline MIC implying the involvement of alternative resistance mechanisms. As CCCP was able to reverse the resistance for tigecycline when used in combination with the tigecycline it confirmed the active role of efflux pumps in tigecycline resistance (Table 2).

REP-PCR

Using 95% similarity cut-off as a threshold, REP-PCR identified 5 clusters (cluster I to V) Cluster II was the major cluster that contained 17 isolates, whereas cluster III is the smallest containing 4 isolates (Fig. S1). Cluster I and IV comprised of 9 and 8 isolates respectively. Nine out of seventeen isolates from cluster II are tigecycline resistant. Cluster III contains three *bla*_{NDM-1} positive isolates and cluster II contains remaining seven *bla*_{NDM-1} positive isolates.

Co-existence of *bla*_{OXA-23} and *bla*_{NDM-1} with *armA*

Carbapenem resistance in *A. baumannii* is generally due to the presence of carbapenemases (Class D β -lactamases) encoding genes, which includes, *bla*_{OXA-23} and *bla*_{OXA-58} along with the intrinsic presence of *bla*_{OXA-51}. In addition, class-A β -lactamases like *bla*_{ADC}, class-B β -lactamases *bla*_{NDM-1} and *bla*_{IMP-1} also enhance the carbapenem resistance towards all the existing carbapenem class of antimicrobial agents in *A. baumannii*. The present study observed a strong co-existence between 16S rRNA methylase and the above-mentioned carbapenem encoding genes wherein we witnessed most of our isolates that carry *bla*_{NDM-1} also harbor the *armA* except for two isolates (ABE33 & ABE61). This finding is in concurrence with the reports from earlier studies wherein the co-existence of carbapenem-resistant determinants in clinical isolates of *A. baumannii* of India and China was documented (Karthikeyan et al. 2010, Chen et al. 2011, Saranathan et al. 2017a). Susceptibility testing showed high level resistance towards both the cephalosporins, ceftriaxone and ceftazidime tested in the study showing the MIC levels of >256 mg/L. All the twenty (10 isolates from this study and 10 isolates used as a reference from our previous collection) (Saranaathan et al. 2017a) isolates were resistant to amikacin, gentamicin, tetracycline and meropenem (Table S2). Interestingly, ten isolates were resistant to tigecycline, were also observed to be carbapenem non-susceptible and such obser-

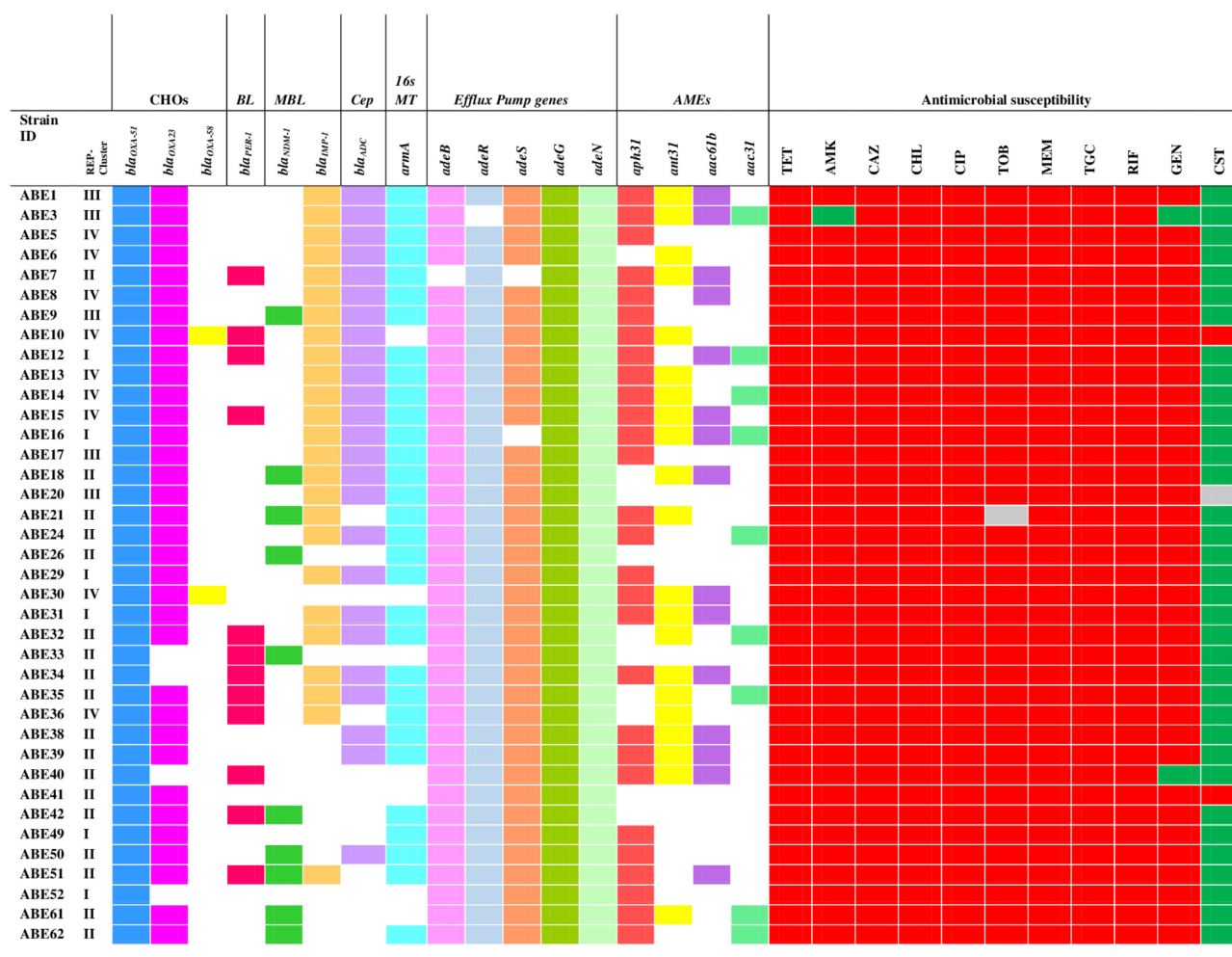


Figure 2. Overview of the antimicrobial resistance determinants, intrinsic β -lactamases encoding genes, aminoglycoside resistance determinants observed in Tigecycline non-susceptible (TNAB) clinical isolates of *A. baumannii*. Colored squares indicate the presence of the gene while the blank cell indicates the absence of the gene. Red squares (Antimicrobial susceptibility) indicate resistance to a specific antibiotic; gray squares, indicate intermediately susceptible to a specific antibiotic; green squares indicate susceptibility to a specific antibiotic. CHO; Carbapenem-Hydrolysing Oxacillinases, MBLs; Metallo- β -lactamase, Cep; Cephalosporinase, BL; Amber class β -lactamase, 16sMT; 16S rRNA methyltransferases, AMEs; Aminoglycoside Modifying Enzymes.

Table 2. Reduction in MIC after addition of efflux pump inhibitor.

Drugs	No. (%) of isolates with 2-fold reduction in MIC		No. (%) of isolates with 4-fold reduction in MIC		No. (%) of isolates with ≥ 8 fold reduction in MIC	
	TNAB (n = 38)	TSAB (n = 13)	TNAB (n = 38)	TSAB (n = 13)	TNAB (n = 38)	TSAB (n = 13)
Tigecycline + CCCP	21 (41.17)	7 (13.72)	11 (21.56)	5 (9.80)	4 (7.84)	1 (1.96)

TNAB, tigecycline-non-susceptible *A. baumannii*; TSAB, tigecycline-susceptible *A. baumannii*; CCCP, carbonyl cyanide 3-chlorophenylhydrazone;

vation has been reported very recently by Lucaßen *et al.* (2021). All these ten isolates were susceptible only to colistin. In addition, PCR screening of OXA type carbapenemase genes revealed that all the isolates harboring *bla*_{NDM-1} also had *bla*_{OXA-23}. PCR was also performed for 16S rRNA methylase-encoding gene *armA*, *rmtC* in the same isolates and it was observed that 18 isolates harbors *armA*, and two of the isolates harbor *rmtC*. Among all the *bla*_{NDM-1} positive isolates, 45% of the isolates were positive for *bla*_{IMP-1}. This is the first study to our knowledge wherein the coexistence of *bla*_{NDM-1}, *bla*_{OXA-51}, *bla*_{OXA-23} and *armA* in eight-tigecycline resistance isolates of *A. baumannii* was observed. The existence of several antibiotic resistance determinants in same strain/single strain limits the

treatment options and the use of combination therapy might not work in such cases.

The presence of multiple resistance genes plays a crucial role in the development of resistance against diverse classes of antimicrobial agents. With respect to the presence of important resistant determinant's such as carbapenemases, aminoglycoside modifying enzymes, 16sMT and efflux pump genes of RND family in the two groups of isolates i. e. TSAB and TNAB, the latter group of isolates show more robust presence of these determinants. Among these isolates, 13 (25.49%) were classified into the TSAB group, and 38 (74.50%) were in the TNAB group. In comparison with TSAB group, TNAB isolates significantly showed increased distribution

of resistant determinants namely *bla*_{OXA-23} ($P = 0.002$), *bla*_{NDM-1} ($P = 0.04$), *bla*_{IMP-1} ($P < 0.0001$), 16sMT gene *armA* ($P = 0.02$) and also efflux pump gene *adeS* ($P = 0.008$) (Table 3).

Sequence analysis of regulatory genes of RND efflux pumps

Analysis of LysR-type transcriptional regulator *adeL*

The *adeL* sequence analysis were done for eight tigecycline resistant strains, the *adeL* sequence of six strains was identical to the reference strain ATCC 17 978. In one of the strains, (ABE51) single Q256R amino acid substitution was found that was earlier reported in MDR *A. baumannii* strain AYE and considered a polymorphism. In another strain (ABE15), we found mutation located in the DNA binding domain (1–53) of *adeL* i. e. G21R is associated with a seven-fold increase in the expression of *adeG* (Table 4). In one of the clinical strains (ABE10), *adeG* expression was more than four-fold higher but did not show any mutation in regulatory gene *adeL*, suggesting that there might be unknown mechanism responsible for overexpression of *adeG* (Fig. 3A)

Analysis of Two component system *adeRS*

The AdeABC is the first RND efflux pump discovered in *A. baumannii* that is regulated by two component system *adeRS* located upstream of AdeABC (Marchand et al. 2004). The two-component system consists of a response regulator *adeR* and a histidine kinase *adeS*. Evidence from earlier studies reveals that amino acid substitutions in *adeR* results in overexpression of *adeB* in tigecycline resistant strains (Yoon et al. 2013). In total sequence analysis of response regulator *adeR* from sixteen tigecycline resistant *A. baumannii* isolates was performed and 14 isolates (Table 4) showed polymorphism at site 136 where alanine was changed to valine. The same fourteen isolates showed mutation V120I located in DNA binding domain (138–247) of *adeR*. An overview of amino acid differences found in efflux pump regulatory genes from tigecycline resistant *A. baumannii* isolates was given in Table 4. We have sequenced *adeS* gene of twenty tigecycline resistant isolates of *A. baumannii*. The SNPs discovered in earlier studies were also found in our isolates and they were listed here: N268H in two isolates (ABE41 and ABE51), V27I in three isolates (ABE21, ABE61 and ABE62), S280A (in ABE21) and Q281D (in ABE21). We did not find the insertion of *ISAbal* in *adeS* in any of our isolate. Mutation N268H has been documented as mere polymorphism in multiple reports (Yoon et al. 2013, Haeili et al. 2021) as it is observed both in tigecycline resistant and susceptible isolates that does not alter the susceptibility status of the isolates. The mutation responsible for overexpression of AdeABC namely A94V reported in an earlier study (Hornsey et al. 2010) was observed in two clinical strains ABE21 and ABE62. Expression analysis of *adeB* in ABE21 showed 18 fold overexpression while in case of ABE62 it was four fold as compared to TSAB strain ATCC 17 978 (Fig. 3B). Interestingly, many novel mutations were observed in clinical isolates such as Q281D, W53R, F12S, F12L, W61R, C70S, and K84E, which were found in multiple TNSAB. Earlier studies have also witnessed mutational changes in all the three major regulatory genes of RND-efflux pumps that are associated with tigecycline resistance (Gerson et al. 2018, Lucaßen et al. 2021 and Xu et al. 2020). However, the consequence of such types of mutations and their exact role in the development of tigecycline resistance is yet to be un-

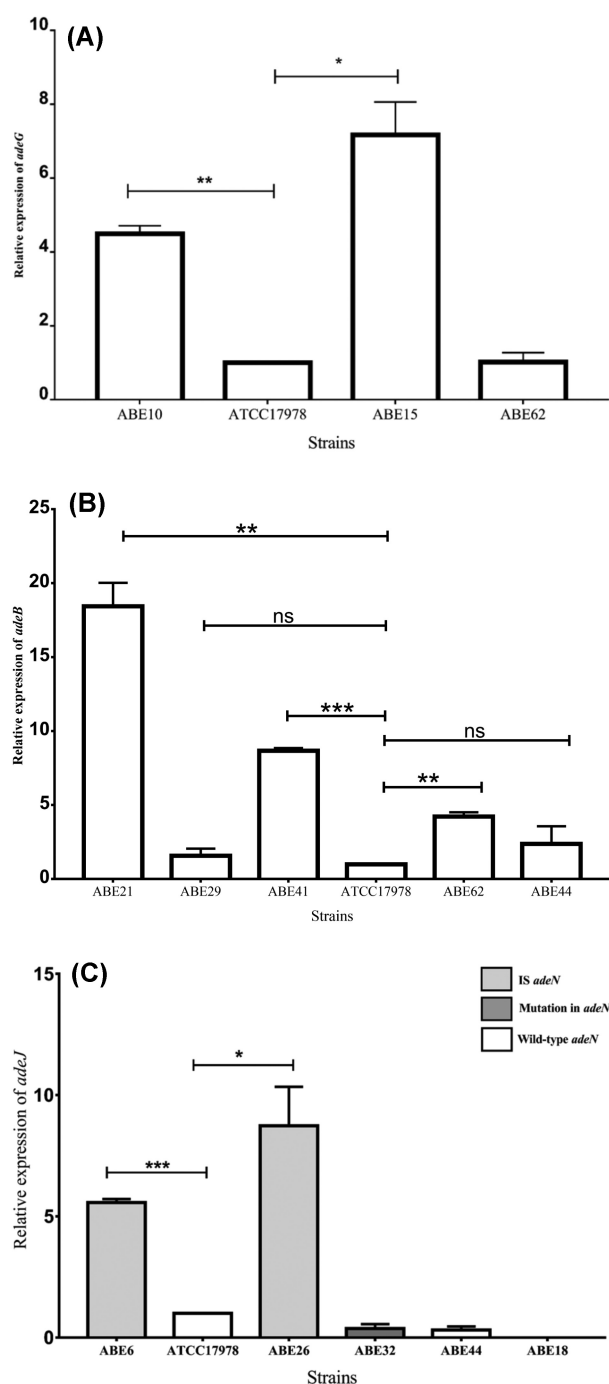


Figure 3. Relative *adeG*, *adeB* and *adeJ* expression in *A. baumannii* clinical isolates determined by qRT-PCR. The Tigecycline-susceptible *Acinetobacter baumannii* ATCC17978 and ABE44 were used for comparison. (A) Expression of *adeG*; (B) Expression of *adeB*; (C) Expression of *adeJ* in *adeN* disrupted TNSAB isolates (ABE6 and ABE26) and substitution mutation containing strains. The number of transcripts in tigecycline-susceptible strains was related to the strains with higher tigecycline MICs after being normalized to the expression of the reference gene *rpoB*. All data are representatives of at least 3 independent experiments. Statistical significance was derived using 2-tailed paired Student's t-test. Error bars represent the standard error (SE) of the mean. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$

Table 3. Distribution of Resistance determinants in 51 clinical isolates of *A. baumannii*.

Gene	TSAB (n = 13)		TNAB (n = 38)		P	Total N (%)
	N	Prevalence rate (%)	N	Prevalence rate (%)		
<i>bla_{OXA-51}</i>	13	100	38	100	–	51 (100)
<i>bla_{OXA-23}</i>	6	46.1	34	89.4	0.0029	40 (78.4)
<i>bla_{OXA-58}</i>	0	0	2	5.2	ns	2 (3.9)
<i>bla_{PER-1}</i>	6	46.1	12	31.5	0.50	18 (35.2)
<i>bla_{NDM-1}</i>	00	00	10	26.31	0.048	10 (19.6)
<i>bla_{IMP-1}</i>	00	00	25	65.7	<0.0001	25 (49)
<i>bla_{ADC}</i>	9	69.2	25	65.7	ns	34 (66.6)
<i>armA</i>	6	46.1	31	81.5	0.0273	37 (72.5)
<i>adeB</i>	10	76.9	37	97.3	0.0464	46 (92.1)
<i>adeS</i>	8	61.5	36	94.7	0.0084	44 (86.2)
<i>adeG</i>	13	100	38	100	–	51 (100)
<i>adeN</i>	12	92.3	38	100	0.2549	50 (98)
<i>aph3</i>	6	46.1	28	73.6	0.0933	34 (66.6)
<i>ant3I</i>	3	23	21	55.2	0.0584	24 (47)
<i>aac6Ib</i>	2	15.3	15	39.4	0.1750	17 (33.3)
<i>aac3I</i>	00	00	9	23.6	0.0900	9 (17.6)

Distribution of Resistance determinants among TSAB and TNAB groups were compared using the Fisher-Exact test, bold values indicate statistical significance.

Table 4. Overview of amino acid mutations found in efflux pump regulatory genes in tigecycline in-resistant *A. baumannii*.

Strain	Site of Isolation	TGC MIC (mg/L)	AdeR	AdeS	AdeN	AdeL	
ABE1	BAL	4	V120I, A136V	G186V	ISAb ^a 1	G21R	
ABE3	ETA	4	V120I, A136V	K84E, G186V			
ABE5	ETA	4	A136V V120I	K84E, W61R			
ABE6	ETA	4					
ABE8	ETA	4	V120I A136V	G186V, L172P			
ABE10	ETA	8	V120I A136V D238H	F12S, G186V, L172P			
ABE13	ETA	4			K67Q		G21R
ABE14	ETA	4	V120I A136V	G186V, L172P			
ABE15	BAL	8					
ABE16	ETA	4	V120I A136V, Q221H	F12S, G186V, L172P			
ABE17	ETA	4	V120I A136V				
ABE18	Wound	4	V120I A136V				
ABE20	Blood	4		G186V, L172P	ISAb ^a 1	G21R	
ABE21	ETA	4		Q281D, A94V, Q299R			
ABE24	Blood	4	V120I A136V				
ABE26	Blood	2					
ABE29	ETA	4	V120I A136V	F12S, G186V, G186V, L172P			
ABE30	Blood	4	Q221H	F12S			
ABE32	Blood	4			G21R		Q256R
ABE34	Blood	4	V120I A136V				
ABE35	Blood	4	V120I A136V				
ABE36	Blood	4		W53R, G186V			
ABE38	ETA	4		F12S, G186V			
ABE39	ETA	4	V120I A136V	L172P			
ABE40	Sputum	4		F12S, G186V, L172P		Q256R	
ABE41	ETA	4		G186V, L172P, N268H			
ABE51	Tissue	4		F12L, G186V, L172P, N268H			
ABE62	Tissue	4		Q281D, Q299R G186V, A94V			

BAL- Bronchiolar alveolar lavage; ETA-Endotracheal Aspirate,

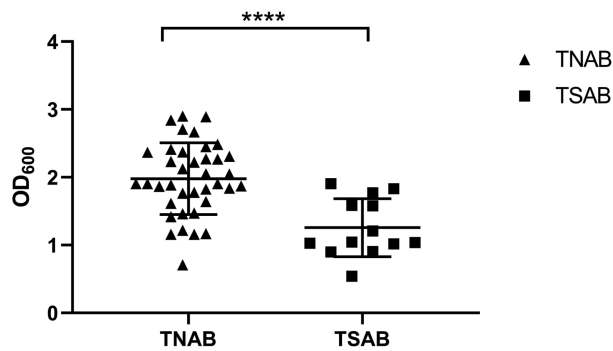


Figure 4. Comparison of biofilm formation capacity between tigecycline non-susceptible (TNAB) and tigecycline susceptible (TSAB) groups of *A. baumannii*. OD data were analyzed by two-tailed t-test ($P < 0.0001$).

Table 5. Tigecycline MIC.

Strains of <i>A. baumannii</i>	Tigecycline MIC (mg/L)
ATCC 17978	0.064
$\Delta adeS$	0.094
$\Delta adeS::pUCAB-adeS$ (CP)	0.094
$\Delta adeS::pUCAB-adeS$ (F12S)	0.094
$\Delta adeR$	0.064

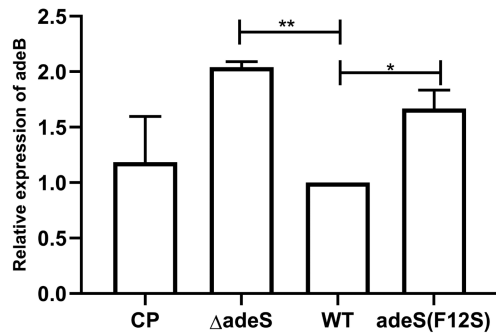


Figure 5. Role F12S mutation in *adeS*. Relative expression of *adeB* in *A. baumannii* ATCC17978 (WT), Complement (CP) *adeS* mutant ($\Delta adeS$) and *adeS* (F12S) determined by qRT-PCR. Statistical significance was derived using 2-tailed paired Student's t-test, ** $P < 0.05$.

derstood clearly. Nevertheless, Yoon *et al.*, (2015) have shown that experimentally induced mutations in isogenic strains modulate the expression levels of RND efflux pump genes that resulted in alterations in membrane associated cellular functions.

Role of SNPs on MIC and expression RND efflux pump

The role of single nucleotide change in the expression of RND efflux pump (*adeABC*) was analyzed by introducing mutation by site-directed mutagenesis. The resultant change in amino acid (F12S) in *adeS* does not alter the MIC towards the tigecycline (Table 5) but we observed a mild change in expression of *adeB* (Fig. 5). The change in expression may not be sufficient to bring a significant change in tigecycline MIC. In addition, we generated *adeR* knockout for furthering the probe. Deletion of *adeR* did not alter the MIC for tigecycline in the deletion mutant. Attempts to create site-directed mutagenesis in *adeR* were failed even after repetitive experiments. However, we found that deletion of *adeS* ($\Delta adeS$)

leads to an increase in the expression of *adeB* (Fig. 5), which is in concurrence with an earlier study in which the tigecycline resistance and mutations in RND efflux pump regulators were analyzed (Gerson *et al.* 2018). This study revealed the association of disruption of *adeS* due to *ISAbal* insertion with an increase in the expression of *adeB*. Nevertheless, in our work, mutation introduced through site-directed mutagenesis in *adeS* was able to show only slight increase in expression of *adeB* (Fig. 5), yet it was not sufficient enough to bring in noticeable change in tigecycline MIC.

Analysis of *tetR* type regulator *adeN*

One of the recent studies, using genomics showed an association between tigecycline resistance and variants of regulators of RND efflux pumps (Gerson *et al.* 2018, Lucaßen *et al.* 2021). The expression level of *adeJ* was used to determine the effect of disruption of *TetR* regulator *adeN* due to *ISAbal* insertion on the expression of *adeIJK* efflux pump as per our earlier investigation (Saranathan *et al.* 2017b). The insertion of *ISAbal* resulted in disruption of *adeN* in two TNSAB isolates namely ABE6 and ABE26 in this study. Sanger sequencing confirmed that the insertion of *ISAbal* in *adeN* was in reverse in orientation. We investigated the effect of this disruption of the regulatory gene on the expression of RND-type efflux pump *adeIJK*. In both these isolates (ABE6 and ABE26), the expression of the RND-type efflux pump was five and eight-fold higher than those isolates that are having intact *adeN* (Fig. 3C). Besides, it has been already shown that the expression of *adeJ* is controlled by *adeN*, where disruption of *adeN* leads to an increase in the expression of *adeJ* and it also increases the pathogenesis and virulence of *A. baumannii* (Rosenfeld *et al.* 2012, Saranathan *et al.* 2017b). Therefore, disruption in *adeN* could be considered an important mutation in *A. baumannii* that enhances resistance. However, the analysis of sequences of *adeN* found two amino acid mutations (K67Q and G21R) in two TNSAB isolates but that did not result in overexpression of efflux pump, inferring the expression of *adeIJK* is tightly controlled by *adeN* and SNPs in *adeN* may not compromise its function (Table 4).

The biofilm-forming potential of isolates was measured using crystal violet staining and optical density measurement, as shown in Fig. 4. TNAB isolates produced biofilm with an OD range from 0.70 to 2.89 while TSAB isolates produced biofilm with an OD range from 0.54 to 1.90, which shows a significant difference in the ability of biofilm formation between tigecycline susceptible and non-susceptible groups ($P < 0.0001$) (Fig. 4). The derived data was analyzed by student's unpaired t-test. The results showed that contemporary *A. baumannii* isolates possess both antibiotic resistance as well as higher ability biofilm formation that is contrary from the observations of earlier studies, which shows that gaining of resistance leads to decrease in bacterial fitness., biofilm formation and virulence potential (López-Rojas *et al.* 2011, Hraiech *et al.* 2013). To the best of our knowledge, this is the first study comparing the biofilm-forming abilities of tigecycline susceptible and non-susceptible clinical isolates of *A. baumannii*.

Conclusion

An increase in tigecycline resistance is a worrisome development in Indian clinical settings and this work is the first study reporting tigecycline resistance in *A. baumannii* from the southern part of the country. Strong correlations that were found between *bla*_{OXA-51}, *bla*_{OXA-23}, *bla*_{NDM-1} and *armA* significantly play a vital role in making bacteria resistant for diverse classes of antimicrobial agents leading to pan-drug resistance. Several amino acid substi-

tutions in *adeRS* and in *adeL* were detected, which might have a role in elevating the expression of *adeB* and *adeG* respectively in TNSAB isolates. The histidine kinase *adeS* has more substitutions than response regulator *adeR* which has been an important finding. The expression analysis of efflux pump *adeIJK* underlines the role of negative regulator *adeN*, wherein its disruption due to *ISAbal* leads to overexpression of the efflux pump (Graphical abstract). Overall, this study gives insight into the diversity and complexity of resistant determinants possessed by clinical isolates of *A. baumannii*. It detected the coexistence of multiple resistant determinants such as *bla_{NDM-1}*, *bla_{OXA-51}*, *bla_{OXA-23}*, and *armA* that are responsible for increased proportions of carbapenem and tigecycline resistance among the strains of *A. baumannii* isolated in India. The study also reconfirms the role of RND-efflux pumps in tigecycline resistance and reveals the increasing diversity of mutations in regulatory genes of efflux pumps. The exact role of novel mutations seen in regulatory genes and consequent development of tigecycline resistance are not understood clearly that has to be deciphered by exhaustive structural and functional genomics.

Supplementary data

Supplementary data are available at [FEMSPD](https://femspd.org) online.

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Competing interests. The authors declare no competing financial interest.

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